

YUXUAN HOU

Shenzhen, China | 12413104@mail.sustech.edu.cn | tsawke.github.io



Education

Southern University of Science and Technology (SUSTech)

Expected Jun 2028

B.Eng. in Computer Science and Technology — Turing Class (Honors Program)

Shenzhen, China

Academic-year GPA: 3.86/4.00 | Class President, 2024 CSE Turing Class

Honors Coursework: Discrete Mathematics; Data Structures & Algorithm Analysis; Algorithm Design & Analysis; Database Systems; Artificial Intelligence.

Selected Coursework: Computer Organization; Computer Vision; Natural Language Processing.

In Progress: Operating Systems (Honors); Compiler Principles; Computer Networks; Advanced Computer Programming; Probability and Statistics for Engineering.

Research Affiliation

Lab: Advanced Intelligent Mobile Systems (AIMS) Lab, SUSTech

Advisor: Jin Zhang, Associate Professor, Department of Computer Science and Engineering

Accepted Publication

Yuxuan Hou, Jiao Li, Linshan Jiang, and Jin Zhang. *Oto-Meal: Earable Sensing with PPG and IMU for Personalized Meal Awareness*.

WellComp 2026 (UbiComp/ISWC 2026 Workshop) — Computing for Well-being.

Accepted; forthcoming in *UbiComp Companion '26*. DOI: 10.1145/3798063.3841767. arXiv:2608.16678.

Additional manuscripts are in preparation or have been submitted for review.

Research Experience

Oto-Meal: Earable Sensing for Personalized Meal Awareness

Sep 2025–Present

Project Lead & Primary Developer

Wearable sensing

Project Lead — 2026 National Undergraduate Innovation and Entrepreneurship Training Program (National Level; CNY 25,000)

Project Lead — 2026 SUSTech “Climbing Program” Project (CNY 15,000; completed)

- Built an audio- and image-free sensing pipeline using photoplethysmography (PPG) and inertial measurement unit (IMU) signals, including preprocessing, event/rest detection, and recognition of five behaviors such as chewing and drinking.
- Organized data from **7 participants, 538 recordings, and 7,028 windows**; implemented reproducible evaluation, modality ablations, and a lightweight user-conditioned memory without retraining the pooled model.
- Achieved **$80.38 \pm 0.84\%$** five-class event accuracy with 20% labeled target-user calibration and **$85.13 \pm 0.57\%$** with 60% under the paper’s window-split protocol; combined PPG+IMU outperformed either modality alone in the reported ablation.
- Led research implementation and first-author manuscript preparation through the accepted camera-ready version; completed the Climbing Program project and continued follow-up experiments on recognition and inference efficiency.

3DGS-Acoustic: Acoustic Reconstruction and Evaluation

May 2026–Present

Research Implementation & Evaluation

Spatial acoustics

- Reproduced DiffRIR on *classroomBase* with 12/155/463 training/validation/test samples and integrated it into a unified room impulse response evaluation framework.
- Implemented dataset and method adapters, checkpoint compatibility, and geometry/input checks; evaluated reconstruction using acoustic decay, spectral, envelope, and amplitude/phase measures.

Ongoing work: Spiking-neural-network sensing, alongside continued acoustic-reconstruction and earable-sensing research.

Selected Research & Engineering Projects

Object-Level Segmentation and Editing in 3D Gaussian Splatting Spring 2026

Custom-Scene Lead | Four-Person Computer Vision Project COLMAP, CUDA

- Led the self-captured-scene workflow, including frame filtering, structure-from-motion reconstruction, pseudo-label generation, Gaussian Grouping training, rendering, and object removal; resolved local GPU extension compatibility.
- Registered **213/213 frames** with a **0.71-pixel** reprojection error and trained a model with approximately **365,000 Gaussians**; achieved a training peak signal-to-noise ratio (PSNR) of 28 dB before object-level editing.

RydbergTrack: Rydberg-Receiver Motion Sensing May–Jun 2026

Signal Processing & Data Analysis | Three-Person Project MATLAB

- Refactored an existing script into a reproducible batch-processing pipeline for **150 million samples** across 15 pre-collected recordings, including complex down-conversion, resampling, phase processing, and Doppler time-frequency analysis.
- Developed a percentile-based phase metric that separated **all 3 static and 12 motion recordings** within the project dataset; produced diagnostic visualizations and qualitative repeated-motion estimates.

Technical Skills

Programming: Python, C/C++, MATLAB, Java, Verilog
Learning & Scientific Computing: PyTorch, scikit-learn, NumPy, SciPy
Systems & Research Tools: CUDA, COLMAP, Vivado, Git, Linux, LaTeX; experimental evaluation, literature review, and bilingual research communication

Selected Honors & Awards

- 2025** **National First Prize**, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)
- 2025** **Silver Medal (Rank 4)**, International Collegiate Programming Contest (ICPC)
- 2025** **Silver Medal**, China Collegiate Programming Contest (CCPC)
- 2025** **Gold Medal**, Guangdong Collegiate Programming Contest
- 2026** **First Prize (1st Place)**, Application Development Division Finals, 1st Hong Kong International AIGC Cultural Digital Content Creation Competition
- 2025** Provincial First Prize, 16th Lanqiao Cup, Guangdong; C/C++ Programming, University Group A
- 2024/25** SUSTech University Scholarship; Zhiren College Motto Scholarship “Rixin”; SUSTech Outstanding Student Leader
- 2024** Undergraduate Talent Development Excellence Program Grant (2024 Cohort; CNY 6,000)
- 2025/26** SUSTech Outstanding Communist Youth League Cadre
- 2026** Second Prize, CCF Service Computing Innovation Competition / Sangfor CoStrict AI-Native Innovation Competition; GeoAgent (team project)

Leadership & Service

President, Student Union | Chair, Student Development Committee

- Lead organizational planning and team coordination, including recruitment, student activities, member development, and cross-organization collaboration; improve operating procedures and leadership handovers.

Head, Zhiren College Peer Mentor Team

- Lead a **15-member team serving 231 incoming students** across 5 classes, with outreach covering 25 provinces; organize onboarding, orientation, and ongoing student support, among other activities.

Collective honors: SUSTech Outstanding Student Union; First Prize, Zhiren College Outstanding Student Organization (2025–2026).

Other roles: Class President, CSE Turing Class; Vice President, 3D Manufacturing Club; member of the university competitive programming, supercomputing, and rowing teams.

Campus service: Fenghua Emcee Team and SUSTech Exhibition Hall Docent Team; former publicity and student-organization roles.